Chapter 6. Cycles in Permutations

Not So Vicious Cycles

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A permutation $p = p_1 p_2 \cdots p_n$ of $[n]$ is also the bijection $p : [n] \to [n]$, $p(i) = p_i$.

Example 6.1

312 is the function f with $f(1) = 3$, $f(2) = 1$, $f(3) = 2$.

The **product** fg of permutations of $[n]$ is the composition "first f, then g": $(fg)(i) = g(f(i))$.

Example 6.2

For $f = 312$, $g = 213$: $(fg)(1) = g(3) = 3$, $(fg)(2) = g(1) = 2$, $(fg)(3) = g(2) = 1$, so $fg = 321$.

Example 6.3

$gf = 132 \neq fg$: multiplication is not commutative.

6.1 Cycles in permutations

Motivation. Every permutation splits into cycles; counting by cycles gives Stirling numbers of the first kind.

In $g = 321564$: 2 is fixed, $1 \leftrightarrow 3$ (a 2-cycle), $4 \to 5 \to 6 \to 4$ (a 3-cycle).

Lemma 6.4

For a permutation p of $[n]$ and $x \in [n]$, there is $1 \leq i \leq n$ with $p^i(x) = x$.

Proof. Otherwise $p(x), \ldots, p^n(x)$ lie in $[n] \setminus \{x\}$, so $p^j(x) = p^k(x)$ with $1 \leq j < k \leq n$ (pigeon-hole), and applying p^{-j} gives $x = p^{k-j}(x)$, $1 \leq k - j \leq n - 1$. ■

Definition 6.5 (Cycle)

Let p be a permutation of $[n]$, $x \in [n]$ and i the smallest positive integer with $p^i(x) = x$. Then $x, p(x), \ldots, p^{i-1}(x)$ form an **i-cycle** of p.

Corollary 6.6

Every permutation is the disjoint union of its cycles.

Proof. Every x lies in a cycle (Lemma 6.4). Cycles sharing z both equal the orbit $\{p^m(z) \mid m \in \mathbb{Z}\}$. ■

Example 6.7

321564 has the cycles (31), (2), (564).

In cycle notation $g(i)$ follows i cyclically; e.g. $(241)(35) = (53)(412)$. The **canonical cycle form** starts each cycle with its maximum and orders cycles by increasing first elements: $(412)(53)$. **One-line notation**: $f(1)f(2)\cdots f(n)$.

Example 6.8

(412)(53) in one-line notation is 24513.

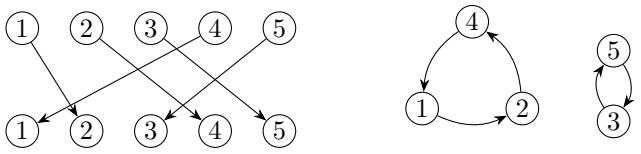


Figure. $24513 = (412)(53)$ *as a function* $[5] \to [5]$ *(left) and as a union of cycles (right).*

Permutations of $[n]$ are called n**-permutations**; their set is the symmetric group S_n.

Theorem 6.9

Let $a_1, \ldots, a_n \geq 0$ with $\sum_i i\, a_i = n$. The number of n-permutations with a_i cycles of length i ($i \in [n]$) is

$$\frac{n!}{a_1!\, a_2! \cdots a_n! \cdot 1^{a_1} 2^{a_2} \cdots n^{a_n}}. \tag{6.1}$$

Proof. Write $[n]$ in a row ($n!$ ways) and bracket a_1 cycles of length 1, then a_2 of length 2, etc. Each permutation of this type arises from $\prod_i i^{a_i} a_i!$ rows (i starting points per i-cycle, $a_i!$ orders of the i-cycles). ■

$(a_1, \ldots, a_n)$ is the (**cycle**) **type** of the permutation.

Example 6.10

There are $(n-1)!$ n-permutations with one cycle (seatings at a round table up to rotation).

Definition 6.11 (Stirling numbers of the first kind)

The number of n-permutations with exactly k cycles is the **signless Stirling number of the first kind** $c(n,k)$; $s(n,k) := (-1)^{n-k} c(n,k)$ is the **Stirling number of the first kind**. $c(n,0) = 0$ for $n > 0$, $c(0,0) = 1$, and $c(n,k) = 0$ for $n < k$.

Theorem 6.12

For positive integers $n \geq k$:

$$c(n,k) = c(n-1, k-1) + (n-1)\, c(n-1,k). \tag{6.2}$$

Proof. Either n is fixed and the rest form $k-1$ cycles ($c(n-1,k-1)$), or the rest form k cycles and n is inserted right after one of the $n-1$ entries ($(n-1)c(n-1,k)$); these choices are distinct and exhaustive. ■

Lemma 6.13

For every positive integer n:

$$\sum_{k=0}^{n} c(n,k) x^k = x(x+1)\cdots(x+n-1). \tag{6.3}$$

Proof. Let $G_n(x) := x(x+1)\cdots(x+n-1) = \sum_k a_{n,k}x^k$ (with $G_0 = 1$). From $G_n = (x+n-1)G_{n-1}$,

$$\sum_k a_{n,k}x^k = \sum_k a_{n-1,k-1}x^k + (n-1)\sum_k a_{n-1,k}x^k.$$

So $a_{n,k}$ satisfies (6.2) with the same initial values, and $a_{n,k} = c(n,k)$ by induction. ■

Replacing x by $-x$ and multiplying by $(-1)^n$:

$$\sum_{k=0}^{n} s(n,k)x^k = (x)_n. \tag{6.4}$$

With (5.2), the lower-triangular matrices $S = (S(n,k))$ and $s = (s(n,k))$ are the change-of-basis matrices between the bases $\{x^k\}$ and $\{(x)_k\}$ of the real polynomials.

Theorem 6.14

The matrices S and s are inverse to each other: $Ss = sS = I$.

Proof. Sums are finite (lower triangular). Substituting (6.4) into (5.2),

$$x^n = \sum_k S(n,k)(x)_k = \sum_j x^j \sum_k S(n,k)s(k,j).$$

By linear independence $\sum_k S(n,k)s(k,j) = \delta_{nj}$: $Ss = I$. Likewise $(x)_n = \sum_j (x)_j \sum_k s(n,k)S(k,j)$ gives $sS = I$. ■

6.2 Permutations with restricted cycle structure

Motivation. Erasing parentheses turns statements about cycles into statements about positions.

Lemma 6.15 (Transition lemma)

For $p \in S_n$ in canonical cycle form, let $g(p)$ be the word obtained by erasing the parentheses, read in one-line notation. Then $g : S_n \to S_n$ is a bijection.

Example 6.16

$g((412)(53)) = 41253$.

Proof of Lemma 6.15. Each $q = q_1 \cdots q_n$ has exactly one bracketing into canonical form: a cycle starts at q_k iff q_k exceeds all earlier leading entries, i.e. (leading entries being cycle maxima) all entries to its left: q_k is a **left-to-right maximum**. These forced brackets give a valid canonical form. ■

Example 6.17

$g^{-1}(4356172) = (43)(5)(61)(72)$: the left-to-right maxima $4, 5, 6, 7$ open the cycles.

t left-to-right maxima of q give t cycles of $g^{-1}(q)$.

Proposition 6.18

For $i \neq j$ in $[n]$, exactly half of all n-permutations contain i and j in the same cycle.

Proof. Relabel so $i = n$, $j = n-1$. In $q = g(p)$, n starts the last cycle, which is n and everything to its right. So they share a cycle iff $n-1$ is right of n in q: half of all q. ■

Lemma 6.19

For $i \in [n]$ and every $k \in [n]$, exactly $(n-1)!$ n-permutations have i in a cycle of length k.

Proof. Let $i = n$. If $q_j = n$ in $q = g(p)$, the cycle of n is $q_j \cdots q_n$, of length k iff $j = n+1-k$: $(n-1)!$ words. ■

So $P(\text{given entry in a } k\text{-cycle}) = \frac{1}{n}$ for every k.

ODD(m), EVEN(m): m-permutations with all cycles of odd, resp. even, length.

Lemma 6.20

$|\text{ODD}(2m)| = |\text{EVEN}(2m)|$ for all positive integers m.

Proof.

Step 1 (the map Φ). $\pi \in \text{ODD}(2m)$ has an even number $2k$ of cycles $C_1, \ldots, C_{2k}$ (canonical order). For $i \leq k$ move the last element of C_{2i-1} to the end of C_{2i} (a singleton disappears): lengths $|C_{2i-1}| - 1$, $|C_{2i}| + 1$, both even. The moved element is at most the leading entry of C_{2i-1}, hence below that of C_{2i}, so canonical form is kept: $\Phi(\pi) \in \text{EVEN}(2m)$.

Step 2 (the inverse). For $\sigma \in \text{EVEN}(2m)$ with cycles $c_1, \ldots, c_h$, work from the right. If the last entry of c_h exceeds the first of c_{h-1}, it was a singleton just before c_h: restore it and continue with c_{h-1}. Otherwise move it back to the end of c_{h-1} and continue with c_{h-2}. A lone remaining cycle's last entry was a singleton. Each step is forced, so the preimage is unique. ■

Example 6.21

$\Phi((4)(513)(726)(8)) = (5134)(72)(86)$.

Example 6.22

$\Phi^{-1}((41)(62)(75)(83)) = (412)(6)(753)(8)$.

Example 6.23

$\Phi^{-1}((21)(53)(64)(87)) = (1)(2)(534)(6)(7)(8)$.

Theorem 6.24

For every positive integer m:

$$|\text{ODD}(2m)| = |\text{EVEN}(2m)| = 1^2 \cdot 3^2 \cdot 5^2 \cdots (2m-1)^2. \tag{6.5}$$

Proof. Count EVEN$(2m)$ (Lemma 6.20). Build p along cycles from 1; when a cycle closes, continue with the smallest unused element.

Step 1. $p(1) \neq 1$: $2m-1$ choices; $p^2(1) \neq p(1)$: $2m-1$ choices (1 closes a 2-cycle).

Step 2. For the $(2i-1)$-st image, $2i-2$ elements are used and the current cycle is even or just starting; the image is neither used nor the start (else an odd cycle closes): $2m-2i+1$ choices. For the $2i$-th image the cycle is odd, so closing is allowed: any of the $2m-(2i-1)$ non-images.

Step 3. $|\mathrm{EVEN}(2m)| = \prod_{i=1}^{m}(2m-2i+1)^2$. ■

For odd n, $\mathrm{EVEN}(n) = \emptyset$, but:

Theorem 6.25

For every positive integer m:

$$|\mathrm{ODD}(2m+1)| = (2m+1)\,|\mathrm{ODD}(2m)| = 1^2 \cdot 3^2 \cdots (2m-1)^2\,(2m+1). \tag{6.6}$$

Proof. By Lemma 6.27 below, Ψ is a bijection $\mathrm{ODD}(2m) \times [2m+1] \to \mathrm{ODD}(2m+1)$; combine with Theorem 6.24. ■

An m-permutation in cycle form has $m+1$ **gap positions**: at the beginning and after each entry.

Example 6.26

$(42)(513)$ has six gap positions: $|(4|2|)(5|1|3|)$.

The map Ψ. For $\pi \in \mathrm{ODD}(2m)$, $k \in [2m+1]$: add 1 to each entry of $\Phi(\pi)$ and insert 1 into the k-th gap (at the front: a new fixed point (1)). One cycle becomes odd and canonical form is kept. Apply Φ^{-1} (relabelled) to the other, even cycles; $\Psi(\pi, k)$ has only odd cycles.

Lemma 6.27

Ψ is a bijection from $\mathrm{ODD}(2m) \times [2m+1]$ onto $\mathrm{ODD}(2m+1)$.

Proof. For $\pi' \in \mathrm{ODD}(2m+1)$: apply Φ to the cycles not containing 1 (an even number of elements); read k as the gap of 1 and delete 1, making its cycle even; subtract 1 and apply Φ^{-1}. Each step inverts one of Ψ. ■

6.3 Selected exercises

Motivation. Book exercises using the cycle decomposition.

Exercise 6.5 (Involutions)

Let $r(n)$ be the number of n-permutations p with $p^2 = \mathrm{id}$, and $r(0) = 1$. Prove that $r(n+1) = r(n) + n\,r(n-1)$ for $n \geq 1$.

Solution. $p^2 = \mathrm{id}$ iff all cycles have length 1 or 2. Split by the cycle of $n+1$.

Step 1 ($n+1$ is fixed). $p|_{[n]}$ is any involution of $[n]$: $r(n)$.

Step 2 ($n+1$ is in a 2-cycle). $(k\ n{+}1)$ for one of n values k, and an involution of $[n]\setminus\{k\}$: $n\,r(n-1)$ ($r(0) = 1$ for $n = 1$). ■

Exercise 6.7 (The average number of cycles)

Prove that n-permutations have on average $H_n = \sum_{i=1}^{n} \frac{1}{i}$ cycles.

Solution.

Step 1 (cycles as a sum over entries). With $\ell_p(i)$ the length of the cycle of i, each cycle contributes $\ell \cdot \frac{1}{\ell} = 1$:

$$\operatorname{cyc}(p) = \sum_{i=1}^{n} \frac{1}{\ell_p(i)}.$$

Step 2 (double counting).

$$\sum_{p \in S_n} \operatorname{cyc}(p) = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{1}{k} \left|\{p : \ell_p(i) = k\}\right| = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{(n-1)!}{k} = n!\, H_n,$$

by Lemma 6.19.

Step 3 (conclusion). Divide by $|S_n| = n!$. ■

Exercise 6.21 (Square roots of permutations)

A permutation $p \in S_n$ has a *square root* if $q^2 = p$ for some $q \in S_n$. Prove that p has a square root iff, for every even m, the number of cycles of p of length m is even.

Solution.

Step 1 (square of one cycle). For $c = (x_0\, x_1 \cdots x_{\ell-1})$, $c^2(x_i) = x_{i+2}$ (indices mod ℓ). For odd ℓ, 2 is invertible mod ℓ, so $c^2 = (x_0\, x_2\, x_4 \cdots)$ is one ℓ-cycle; for even ℓ, $c^2 = (x_0\, x_2 \cdots x_{\ell-2})(x_1\, x_3 \cdots x_{\ell-1})$, two $\ell/2$-cycles.

Step 2 (necessity). If $q^2 = p$, p is the union of the squares of the cycles of q (Step 1). An even m-cycle of p comes from a $2m$-cycle of q, which yields two m-cycles; so m-cycles come in pairs.

Step 3 (sufficiency). Pair the cycles of each even length. For an odd cycle c of length $2r+1$, $q = c^{r+1}$ gives $q^2 = c^{2r+2} = c$. For a pair $(y_0 \cdots y_{m-1})$, $(z_0 \cdots z_{m-1})$, $q = (y_0\, z_0\, y_1\, z_1 \cdots y_{m-1}\, z_{m-1})$ gives $q^2(y_i) = y_{i+1}$, $q^2(z_i) = z_{i+1}$.

Step 4 (conclusion). These pieces live on disjoint sets covering $[n]$, so they define $q \in S_n$ with $q^2 = p$. ■